

1 Basics

- Taylor: $f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$
 - $f(a) + \frac{\partial f(x)}{\partial x} \Big|_a - \frac{1}{2}(x-a)^\top \left(\frac{\partial^2 f(x)}{\partial x \partial x^\top} \right) \Big|_a (x-a)$
 - Power series of exp.: $\exp(x) := \sum_{k=0}^\infty \frac{x^k}{k!}$
- Entropy: $H(X) = \mathbb{E}_X[-\log \mathbb{P}(X=x)]$
- Diverg.: $D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \geq 0$
- $1-z \leq \exp(-z)$
- Cauchy-Schwarz: $|\mathbb{E}[X,Y]|^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2]$
- Jensen, $f(X)$ convex: $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$
- Tot exp: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$
- Raylength: $M = M^\top, R(M,x) = \frac{x^\top M x}{x^\top x} \in [\lambda_{\min}(M), \lambda_{\max}(M)]$

1.1 Probability / Statistics

- Gaussian: $\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}\right)$
 $\mathcal{N}(x|\mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2} (x-\mu)^\top \Sigma^{-1} (x-\mu)\right)$
 $X \sim \mathcal{N}(\mu, \Sigma), Y = A + BX \Rightarrow Y \sim \mathcal{N}(A + B\mu, B\Sigma B^\top)$
- Binom.: $f(k,n;p) = \mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$
- $\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
 $\mathbb{V}[X+Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\text{Cov}(X,Y)$
- $\text{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$
 $= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$
 $\text{Cov}(aX, bY) = ab \text{Cov}(X,Y)$

1.2 Calculus

- $\int uv' dx = uv - \int u'v dx$ • $\frac{\partial}{\partial x} \frac{g}{h} = \frac{g'h - gh'}{h^2}$
- $\frac{\partial}{\partial x} (b^\top A x) = A^\top b$ • $\frac{\partial}{\partial x} (b^\top x) = \frac{\partial}{\partial x} (x^\top b) = b$
- $\frac{\partial}{\partial X} (c^\top X^\top b) = bc^\top$ • $\frac{\partial}{\partial X} (c^\top X b) = cb^\top$
- $\frac{\partial}{\partial x} (x^\top A x) = (A^\top + A)x \stackrel{A \text{ sym.}}{=} 2Ax$
- $\frac{\partial}{\partial X} \text{Tr}(X^\top A) = A$ • Tr. trick: $x^\top A x \stackrel{\text{inner prod.}}{=} \text{Tr}(x x^\top A) \stackrel{\text{cyclic perm.}}{=} \text{Tr}(A x x^\top)$
- $|X^{-1}| = |X|^{-1}$ • $\frac{\partial}{\partial X} \log|X| = X^{-\top}$ • $\frac{d}{dx} |x| = \frac{x}{|x|}$
- $\frac{\partial}{\partial x} \|x\|_2 = \frac{\partial}{\partial x} (x^\top x) = 2x$ • $\frac{\partial}{\partial x} \|x-b\|_2 = \frac{x-b}{\|x-b\|_2}$
- $\frac{\partial}{\partial x} \|x\|_1 = \text{sgn}(x)$ $\text{sgn}(x) \in \{\pm 1\}^p$ is row-wise
- $\sigma(x) = \frac{1}{1+\exp(-x)} \Rightarrow \nabla \sigma(x) = \sigma(x)(1-\sigma(x))$
- $\tanh x = \frac{2\sinh x}{2\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ • $\nabla \tanh x = 1 - \tanh^2 x$

1.3 PCA

Data matrix $X \in \mathbb{R}^{n \times d}$, mean-centered: $\tilde{X} = X - \mathbf{1}_n \mu^\top$, $\mu = \frac{1}{n} \sum_i X_i$. Cov. matrix: $\Sigma = \frac{1}{n} \tilde{X}^\top \tilde{X}$. Eigendecomp.: $\Sigma = Q \Lambda Q^\top$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_1 \geq \dots \geq \lambda_d \geq 0$. If $M = D - 1$ error is $J = q_D^\top \Sigma q_D$, q_D eig.vec. smallest eig.val. λ_D .

2 Density Estimation

Bayesianism: Prior $P(\theta)$, likelihood $P(X|\theta)$, posterior $P(\theta|x_{1:n})$. **Bayes:** $P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}$
Frequentism: Param. model $P(Y|X,\theta)$, likelihood $P(X,Y|\theta)$. $\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} P(X,Y|\theta)$.
Ex. Gaussian: $X_{i=1}^n \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow \log \mathcal{L}_n(\mu, \sigma) = -n \log \sigma - \frac{nS^2}{2\sigma^2} - \frac{n(\bar{X} - \mu)^2}{2\sigma^2} \Rightarrow \hat{\mu} = \bar{X}, \hat{\sigma} = S$

2.1 Estimation - MLE Properties

Consistency: $\forall \epsilon > 0, \mathbb{P}\{|\hat{\theta}_n - \theta^*| > \epsilon\} \xrightarrow{n \rightarrow \infty} 0$
Equivariance: If $\hat{\theta}_n$ is MLE of θ , then $g(\hat{\theta}_n)$ is MLE of $g(\theta)$.
Asympt. normality: $\frac{1}{\sqrt{n}}(\hat{\theta}_n - \theta^*) \rightarrow \mathcal{N}(0, J^{-1}(\theta^*) I_n(\theta^*) J^{-1}(\theta^*))$
Asympt. efficiency: $\hat{\theta}_n$ minimises $\mathbb{E}[(\hat{\theta}_n - \theta^*)^2]$ as $n \rightarrow \infty$, i.e. $\mathbb{E}[(\hat{\theta}_n - \theta^*)^2] \stackrel{n \rightarrow \infty}{=} \frac{1}{I_n(\theta^*)}$ (Rao Cr.)
Among all consistent estimators $\hat{\theta}_n$ has *smallest variance*: $\lim_{n \rightarrow \infty} (\mathbb{V}[\hat{\theta}_n] I_n(\theta^*))^{-1} = 1$

2.2 Rao Cramer inequality all $\mathbb{E}$ w.r.t. $P(x|\theta^*)$

Score func.: $\Lambda = \frac{\partial \log \mathbb{P}(x|\theta)}{\partial \theta}$, $\mathbb{E}[\Lambda] = 0$
Fisher info.: $I_n(\theta) = \mathbb{V}[\Lambda]$
 $J(\theta) = \mathbb{E}[\Lambda^2] = -\mathbb{E} \left[\frac{\partial^2 \log \mathbb{P}(x|\theta)}{\partial \theta \partial \theta^\top} \right] = -\mathbb{E} \left[\frac{\partial \Lambda}{\partial \theta} \right]$

General bound: $\mathbb{E}[(\hat{\theta}_n - \theta^*)^2] \geq \frac{(1 + \frac{\partial}{\partial \theta} b_\theta)^2}{\mathbb{E}[\Lambda^2]} + b_\theta^2$

Unbiased case: $\mathbb{E}[(\hat{\theta}_n - \theta^*)^2] = \mathbb{V}[\hat{\theta}_n] \geq \frac{1}{I_n(\theta^*)}$

Tradeoff: $\mathbb{E}[(\hat{\theta}_n - \theta^*)^2] = \mathbb{V}[\hat{\theta}_n] + \text{bias}^2(\hat{\theta}_n)$

Bias: $\text{bias}(\hat{\theta}_n) \equiv b_\theta(\theta^*) = \mathbb{E}[\hat{\theta}_n] - \theta^* \stackrel{\text{unbiased}}{=} 0$

3 (Linear) Regression model: $\hat{y} = X\beta$

Assuming $X^\top X$ non-singular.
Bayesian view: $(Y|X, \beta) \sim \mathcal{N}(X^\top \beta, \sigma^2 \mathbb{I})$
Distrib. of estimator $\hat{\beta}_{\text{LS}} \sim \mathcal{N}(\beta, (X^\top X)^{-1} \sigma^2)$
Ridge: $\epsilon_{\text{RSS}}(\beta, \lambda) = (y - X^\top \beta)^\top (y - X^\top \beta) + \lambda \beta^\top \beta$
 $\hat{\beta} = (X^\top X + \lambda \mathbb{I})^{-1} X^\top y$, prior: $\beta \sim \mathcal{N}(0, \frac{\sigma^2}{\lambda} \mathbb{I})$
(Ridge) Shrinkage: Decompose $X = U D V^\top$

$$X \hat{\beta} = U D (D^2 + \lambda \mathbb{I})^{-1} D U^\top y = \sum_{j \leq d} u_j \frac{d_j^2}{d_j^2 + \lambda} u_j^\top y$$

Lasso: $\hat{\beta} = \arg\min_{\beta} \sum_{i \leq n} (y_i - x_i^\top \beta)^2 + \lambda \|\beta\|_1$
(no closed form), prior: $p(\beta_i) = \frac{\lambda}{4\sigma^2} \exp(-|\beta_i| \frac{\lambda}{2\sigma^2})$

Bias-variance: $\text{MSE} = \mathbb{E}_{D,X,Y}[(\hat{f}(x) - Y)^2] = (\mathbb{E}_{D,X}[\hat{f}(x)] - \mathbb{E}_Y[Y])^2 + \mathbb{E}_D[(\mathbb{E}_{D,X}[\hat{f}(x)] - \hat{f}(x))^2] + \mathbb{E}[(\mathbb{E}_Y[Y] - Y)^2] = \text{bias}^2 + \text{variance} + \text{noise}$

Gauss-Markov Theorem: For any linear estimator $\tilde{\theta} = c^\top y = a^\top (\hat{\beta} + D y)$ that is unbiased for

$a^\top \beta$, it holds: $\mathbb{V}[a^\top \hat{\beta}] \leq \mathbb{V}[c^\top y]$. Among all linear u-estimators, $\hat{\beta}_{\text{LS}}$ minimises the gen. error!

3.1 3.1 Causality

Shortcut Learning: Model uses spurious correlation $X \leftrightarrow Y$ (e.g. grass $\leftrightarrow$ cow)
 $X \overset{\perp}{\dashv} A$: Part of X **not** causally influenced by A
Causal: $X \overset{\perp}{\dashv} W$ and $X, W \& Y$ are causes of Y .
Anti-Causal: $X \overset{\perp}{\dashv} W$ and $X, W \& Y$ are caused by Y .
 W non-causal feature (e.g. background) influences X but not Y . **Counterfactually Inv.:** $f(X(w)) = f(X(w'))$ for $w, w' \in W$. **Confounding:** Hidden U affects W, X . **Selection:** Hidden S filters on W, X . **Theorem For Counterfactual Invariance (CI):** Anti-causal: $f(X) \perp W|Y$ Causal (no selection): $f(X) \perp W$ Causal (no confounding): $Y \perp X|X^\perp w, W$ and $f(X) \perp W|Y$.

4 Uncertainty Quantification

4.1 Bayesian Linear Regression

Model: $y = X^\top \beta + \epsilon$, with $\epsilon \sim \mathcal{N}(\epsilon|0, \sigma^2 \mathbb{I})$
Likelihood: $P(y|X, \beta, \sigma) = \mathcal{N}(y|X^\top \beta, \sigma^2 \mathbb{I})$
Prior: $P(\beta|\Lambda) = \mathcal{N}_d(\beta|0, \Lambda^{-1})$ (Ridge regr. if $\Lambda = \lambda \mathbb{I}$ and $\sigma = 1$)
Posterior: $P(\beta|X, y, \Lambda) = \mathcal{N}(\beta|\mu_\beta, \Sigma_\beta)$
with $\mu_\beta = (X^\top X + \sigma^2 \Lambda)^{-1} X^\top y$
and $\Sigma_\beta = \sigma^2 (X^\top X + \sigma^2 \Lambda)^{-1}$
 $\Rightarrow y \sim \mathcal{N}(y|0, X \Lambda^{-1} X^\top + \sigma^2 \mathbb{I})$ using $\mathbb{E}_{\beta, \epsilon}[\cdot]$
kernel $k(x_i, x_j) := x_i^\top \Lambda^{-1} x_j$

4.2 Gaussian Process

$y \sim \mathcal{N}(y|m(X), K(X,X) + \sigma^2 \mathbb{I})$
 $\begin{bmatrix} y \\ y_{n+1} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} y \\ y_{n+1} \end{bmatrix} \middle| \begin{bmatrix} m(X) \\ m(x_{n+1}) \end{bmatrix}, \begin{bmatrix} C_n & k \\ k^\top & c \end{bmatrix} \right)$
 $p(y_{n+1}|x_{n+1}, X, y) = \mathcal{N}(y_{n+1}|\mu_{n+1}, \sigma_{n+1}^2)$
with $\mu_{n+1} = m(x_{n+1}) + k^\top C_n^{-1} (y - m(X))$
and $\sigma_{n+1}^2 = c - k^\top C_n^{-1} k$
where $K = k(X,X)$, $k = k(x_{n+1}, X)$,
 $C_n = K + \sigma^2 \mathbb{I}$, $c = k(x_{n+1}, x_{n+1}) + \sigma^2$

4.3 Kernels scalar product $K_{ij} = k(x_i, x_j)$

Valid kernel: must be **symmetric** and **p.s.d.** ($x^\top K x \geq 0 \forall x$ or pos. eigenvalues or pos. principal minors). Must have a (pot. ∞ -dim.) **feature vector** ϕ s.t. $k(x, x') = \phi(x)^\top \phi(x')$.
Common kernels: Linear: $x^\top x'$. Polynomial: $(x^\top x' + 1)^p$, $p \in \mathbb{N}$. RBF (Gaussian): $\exp(-\|x - x'\|_2^2 / h^2)$. Sigmoid: $\tanh(\kappa \cdot x^\top x' - b)$.
Kernel construction: • $k_1 + k_2$ • $c \cdot k_1$, $c > 0$
• $k_1 \cdot k_2$ • $f(x) k_1(x, x') f(x')$
• $k(\phi(x), \phi(x'))$ with $\phi: \mathcal{X} \rightarrow \mathbb{R}^d$
• $g(k_1)$ with g : exp. or polyn. w/ all pos. coeff.

4.4 Active Learning $I(X;Y) = H(X) - H(X|Y)$

Mutual Info: $I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$
Transd.-L.: $x_n = \arg\max_{x \in \mathcal{S}} I(f_{x \in A}; y_{x \in A} | D_{n-1})$
 $\rightarrow$ For $f \sim \text{GP}(\mu, k)$: $I = \frac{1}{2} \log \frac{\text{Var}(y_{x \in A} | D_{n-1})}{\text{Var}(y_{x \in A})}$
Safe BO: GP s_{f^*}, g^* observe $y_i = f^*(x_i), z_i = g^*(x_i)$

95% conf.: $[l_n^f, u_n^f], [l_n^g, u_n^g]$
 $S_n = \{x: l_n^g(x) \geq 0\}, \hat{S}_n = \{x: u_n^g(x) \geq 0\}$
 $A_n = \{x \in \hat{S}_n: u_n^f(x) \geq \max_{x' \in S_n} l_n^f(x')\}$
Now ITL on sample space S_n , target space A_n .
Batch Active Learning: $B_\delta(x) = \{x': \|x - x'\| \leq \delta\}$
 $\Rightarrow \arg\max_{L \subseteq X, |L|=b} \Pr(\cup_{x \in L} B_\delta(x))$
 $\Rightarrow \arg\max_{L \subseteq X, |L|=b} \frac{1}{|X|} |\{x': \|x' - x\| \leq \delta, x \in L\}|$
NP-hard Problem, greedy algo (ProbCover):
 $L = \emptyset, E = \{(x, x'): \|x - x'\| \leq \delta\}$ For $i = 1, \dots, b$:
Add $x_i = \arg\max_{x \in X} |\{x': (x, x') \in E, x' \in X\}|$
Update $L \leftarrow L \cup \{x_i\}, E \leftarrow E \setminus (\{x_i\} \times (B_\delta(x_i) \cap X))$

5 Support Vector Machine (SVM)

Primal (soft margin): $\min_{w, w_0, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$
s.t. $y_i(w^\top x_i + w_0) \geq 1 - \xi_i$ and $\xi_i \geq 0$
 $\hookrightarrow$ intractable if $\varphi(x_i)$ instead of x_i
 $\hookrightarrow \xi_i = 0$ means x_i was not neglected
Dual: $\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{(i,j)} \alpha_i \alpha_j y_i y_j x_i^\top x_j$
s.t. $0 \leq \alpha_i \leq C$; $\sum_i \alpha_i y_i = 0$
 $\hookrightarrow$ solve α via quadratic optimisation
Optimal hyperplane: $w^* = \sum_i \alpha_i^* y_i x_i$
 $\hookrightarrow \alpha_i^* \neq 0$ only for *support vectors*
Optimal slack: $\xi_i^* = \max(0, 1 - y_i(w^{*\top} x_i + w_0^*))$

5.1 Structural SVMs

$\min_{w, \xi} \frac{1}{2} \|w\|^2 + \frac{C}{n} \sum_{i \leq n} \xi_i$ s.t. $\xi_i \geq 0$ and $\forall y' \neq y_i$:
 $f_w(x_i, y_i) \geq \Delta(y_i, y') + f_w(x_i, y') - \xi_i - \epsilon$
 Ψ : joint-feature map; Δ : loss / class dissimilarity func.; $f_w(x, y) = w^\top \Psi(x, y)$: compatibility score btw. x and y ; ϵ : tol. / univer. slack variable.
Prediction: $c(x) = \arg\max_y f_w(x, y)$
Note: For optimal w^*, ξ^* , **emp. risk** $(w^*) \leq \frac{1}{n} \sum_i \xi_i^*$
Training: Start without any constraints. In each iteration, add for each (x_i, y_i) the constraint with $y' \neq y_i$ that is the “most violated” and solve again with quadr. optimisation.

6 Ensemble Methods $\hat{f}(x) \triangleq \frac{1}{B} \sum_{i \leq B} \hat{f}_i(x)$

$\text{bias}[\hat{f}(x)] = \frac{1}{B} \sum \text{bias}[\hat{f}_i(x)]$
 $\mathbb{V}[\hat{f}] = \frac{1}{B^2} \sum \mathbb{V}_D[\hat{f}_i] + \frac{1}{B^2} \sum \sum_{i,j} \text{Cov}(\hat{f}_i, \hat{f}_j) \approx \frac{\sigma^2}{B}$

6.1 Bagging (Bootstrap aggregation)

1. Draw M bootstrap sets $Z_1', \dots, Z_M'$
2. Train M base models $b^{(1)}, \dots, b^{(M)}$
3. **Aggregate:** $\hat{b}^{(M)}(x) = \begin{cases} \frac{1}{M} \sum_{t \leq M} b^{(t)}(x) & \text{regr.} \\ \text{sign}(\sum_t b^{(t)}(x)) & \text{class.} \end{cases}$

Why it works: Small *variance* (weak learners)

Random Forest: At each splitting step, u.a.r. choose m of p features & split only one (best) feature. $\rightarrow$ reduce *correlation* between trees.

6.2 Boosting

Sequentially train weak learners on all data, but $\nearrow$ weight of misclass. samples ($\searrow$ bias).

AdaBoost: Stat. learning (*forward stagewise additive modeling*) with **exp. loss**, trains max-margin ($= y_i \hat{b}(x_i)$), self-avg. and interpolating ($\searrow$ overfitting) classifiers.

[Init]: $w_i \leftarrow 1/n \forall i \leq n$ for $b = 1 \dots B$:
 [Train]: Fit classifier $c_b(x)$ using weights w_i
 [Eval]: $\epsilon_b = \frac{\sum_{i=1}^n w_i \mathbb{I}[c_b(x_i) \neq y_i]}{\sum_{i=1}^n w_i}$ // error of c_b
 [Update]: $\alpha_b = \log(\frac{1-\epsilon_b}{\epsilon_b})$ // always > 0 , final weight of c_b
 [Reweight]: $w_i \leftarrow w_i \exp(\alpha_b \mathbb{I}[c_b(x_i) \neq y_i])$ normalize!
 Return $\hat{c}_{\text{ADA}}(x) = \text{sign}(\sum_{b=1}^B \alpha_b c_b(x))$

7 Deep Learning

Sigmoid: $\sigma(x) = \frac{1}{1+\exp(-x)} = \frac{e^x}{e^x+1}$
 $\sigma'(x) = \sigma(x)(1-\sigma(x)) = \sigma(x)\sigma(-x)$

Softmax: $y_i \propto \exp(\beta z_i)$

7.1 Variational Autoencoders

Def: $\frac{p_{\theta'}(z)}{\text{prior}} \xrightarrow[\text{likelihood}]{\text{dec}_{\theta}(z)=p_{\theta}(z|x)} \mathcal{X} \xrightarrow[\text{approx. posterior}]{\text{enc}_{\theta'}(x)=q_{\phi}(z|x)} \mathcal{Z}$
 sample/obs. $\mathcal{X}_i$ from latent representation $\mathcal{Z}$

Train: $\max_{\theta', \theta, \phi} \sum_i \log p_{\theta', \theta}(x_i)$ (*) indep. of $\mathcal{Z}$

(*) $= \mathbb{E}_{Z \sim q_{\phi}(\cdot|x_i)} \left[\log \left(\frac{p_{\theta', \theta}(x_i, Z)}{p_{\theta', \theta}(x_i)} \frac{q_{\phi}(Z|x_i)}{q_{\phi}(Z)} \right) \right]$
 $\frac{\mathcal{L}(x_i, \theta, \phi)}{\text{ELBO} = \text{Infomax} - \text{Regularisation term}}$
 $= \mathbb{E}[\log p_{\theta}(x_i|Z)] - D_{\text{KL}}(q_{\phi}(\cdot|x_i) \| p_{\theta'}(\cdot))$
 $+ D_{\text{KL}}(q_{\phi}(\cdot|x_i) \| p_{\theta', \theta}(\cdot|x_i)) \geq \mathcal{L}(x_i, \theta, \phi)$

Train: $\theta^*, \phi^* = \arg\max_{\theta, \phi} \mathcal{L}(x_i, \theta, \phi)$

Requirements for good representation:

- **informative:** $\theta^* = \arg\max_{\theta} I(X;Z)$
 $= \arg\max_{\theta} \mathbb{E}_{X,Z} [\log p(X|Z)] - \text{const}_{w.r.t. \theta}$
 $\approx \arg\max_{\theta} \sum_i \mathbb{E}_{Z|x} [\log p(x_i|Z)]$
- **disentangled:** components in $\mathcal{Z}$ associated with distinct feature in $\mathcal{X}$ (see D_{KL} in ELBO).
- **robust:** noise in $\mathcal{Z}$ doesn't substantially affect $\mathcal{X}$ (and vice versa). $\rightarrow$ choice of approx. post.!

7.2 Transformers

Attention: Query: $Q = EW^Q$, Key: $K = EW^K$, Value: $V = EW^V$, embedding E . $A = \text{softmax}(\frac{QK^T}{\sqrt{d}})V$, d : E-dim.

- **Self-Attention:** Q, K, V from same sequence
- **Cross-Att.:** Q from target, K, V from source.
- **Multi-headed:** Multiple weights, Stack A_i 's.
- **Masked Self-Attention:** Future tokens masked ($-\infty$ in upper triangle of $P = QK^T$)

Pos. Encoding: $p_{i,2j} = \sin(\alpha^{-2j}i)$, $p_{i,2j+1} = \cos(\alpha^{-2j}i)$, for $j \leq d/2$, $\alpha = 10^{4/d}$.

7.3 Stable Diffusion

Divergence: $\nabla_x \cdot f(x) = \sum_i \frac{\partial}{\partial x_i} f_i(x) = \text{tr}(J_f(x))$

Brownian Motion: 1) Initial condition $W(0) = 0$
 2) Independent increments 3) $W(t+\Delta t) - W(t) \sim \mathcal{N}(0, \Delta t)$ 4) Continuous paths. **NCSM:** $\mathcal{J}_{\text{NCSM}}(\theta) = \frac{1}{2} \mathbb{E}_{X, \epsilon} [\|s_{\theta}(X+\epsilon, \sigma) + \frac{\epsilon}{\sigma^2}\|^2]$

Langevin Sampling: $X \leftarrow X + \alpha_j s_{\theta}(X, \sigma_j) + \sqrt{2\alpha_j} \epsilon$, $\alpha_j = \alpha \frac{\sigma_j^2}{\sigma_K^2}$. **Hyvärinen's trick:** $\int_{\mathbb{R}^d} s_{\theta}(X) \cdot \nabla_x P(X) dX = - \int_{\mathbb{R}^d} P(X) \nabla_x \cdot s_{\theta}(X) dX$

Diffusion: **Forward Process:** For $t \leq T$, $\beta_t > 0$, let $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$, $q(x_t|x_0) = \mathcal{N}(x_t | \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I)$
Reverse Process: UNet predicts noise $\epsilon_{\theta}(x_t, t)$ added to form x_t from x_0 . Then reconstruct x_{t-1} :
 $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z$, $z \sim \mathcal{N}(0, I)$

7.4 Reinforcement Learning

Rew.-to-go: $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k} = r_t + \gamma G_{t+1}$.

Markov Property: $P(s_{t+1}|s_0:t, a_0:t) = P(s_{t+1}|s_t, a_t)$. **Values:** $V^{\pi}(s) = \mathbb{E}_{\pi} [G_t | s_t = s]$, $Q^{\pi}(s, a) = \mathbb{E}_{\pi} [G_t | s_t = s, a_t = a]$. **Bellman Exp.:** $V^{\pi}(s) = \sum_{a, s'} \pi(a|s) p(s'|s, a) (r + \gamma V^{\pi}(s'))$. **"Opt.:** $Q^*(s, a) = \sum_{s'} p(s'|s, a) (r + \gamma \max_{a'} Q^*(s', a'))$. **Q-Learning:** $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$ (Off-policy). **DQN:** MLP $Q_{\theta} \approx Q^*$. Loss $L(\theta) = (r + \gamma \max_{a'} Q_{\theta} - (s', a') - Q_{\theta}(s, a))^2$. Uses **Replay Buffer** (decorrelates data) and **Target Network** θ^- (stabilises targets). 1) Init replay buff and net params θ, θ^- 2) For each episode 2.1) Observe s_0 2.2) For each timestep 2.2.1) Select action using ϵ -Greedy on $Q_{\theta}(s_t, \cdot)$ 2.2.2) Exec action, observe r_t, s_{t+1} 2.2.3) Store (s_t, a_t, r_t, s_{t+1}) in buff 2.2.4) Sample minibatch from buff 2.2.5) Compute targets using θ^- and update θ by GD on $L(\theta)$ 2.2.6) Every C steps set $\theta^- \leftarrow \theta$. **Policy Gradient:** $\nabla_{\theta} J(\theta) = \mathbb{E}[\sum_i \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t]$. **REINFORCE:** 1) Initialize env. 2) For each epoch 2.1)

Initialize buff. 2.2) For each episode: collect data using π_{θ} , compute G_t , store in buff. 2.3) Compute $\nabla_{\theta} L_{PG}(\theta) = -\frac{1}{|D|} \sum_D \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t$ using buff. 2.4) Update θ using $\nabla_{\theta} L_{PG}(\theta)$. **Actor-Critic:** Actor π_{θ} and Critic V_{ϕ} . Advantage $A_t = G_t - V_{\phi}(s_t)$. **GAE:** $A_t^{GAE} = \sum_{l=0}^{T-t-1} (\gamma \lambda)^l \delta_{t+l}$ where $\delta_t = r_t + \gamma V_{\phi}(s_{t+1}) - V_{\phi}(s_t)$. TD high bias, MC high variance. **AC+GAE:** 1) Initialize env., actor, critic 2-2.2) Same 2.3) For each critic update compute critic loss and optimize it. 2.4) Compute values for each timestep using critic. 2.5) Compute GAE. 2.6) Compute actor loss and optimize it (policy). **PPO:** $L_{CLIP}(\theta) = -\mathbb{E}[\min(w_t A_t, \text{clip}(w_t, 1 - \epsilon, 1 + \epsilon) A_t)]$, $w_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta, \text{old}}(a_t | s_t)}$. 1-2.2) Same as AC+GAE 2-3) 2.4-2.5) Same as AC+GAE 2.6) For each update 2.6.1) Compute critic loss and update it 2.6.2) Same with actor. **Entropy Reg.:** $L_{\text{total}} = L_{\text{policy}} + c_V L_V - c_H H(\pi_{\theta})$ (prevents premature collapse).

8 Clustering

k-means: $\arg\min_{\theta} \sum_{i \leq n} \|x_i - \theta_{c(x_i)}\|^2$

8.1 Mixture Models

Assume: $x \sim p(x | \pi_{1...k}, \theta_{1...k}) = \sum_{c \leq k} \pi_c p(x | \theta_c)$

Find: $\hat{\theta} = \arg\max_{\theta} p(\mathcal{X} | \pi, \theta) = \prod_x p(x | \pi, \theta)$

Gaussian Mixtures: $\rightarrow p(x | \theta_c) = p(x | \mu, \Sigma)$
 Introduce *latent indicator variables* for mode assignments $M_{x,c} \in \{0, 1\}$. Then, the **log-likelihood:** $L(\mathcal{X}, M | \theta) = \sum_x \sum_{c \leq k} M_{x,c} \log(\pi_c p(x | \theta_c))$

8.2 Non-parametric Bayesian Methods

$\text{Dir}(x | \alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^n x_k^{\alpha_k - 1}$, $B(\alpha) = \frac{\prod_{k=1}^n \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^n \alpha_k)}$

Rewrite **Finite mixture models:**
 $p(x) = \sum_{k=1}^K \pi_k p(x | \theta_k) = \int p(x | \theta) G(\theta) d\theta$
 where $G(\theta) = \sum_{k=1}^K \pi_k \delta_{\theta_k}(\theta) \leftarrow$ discrete distr.

8.2.1 Finite GMM

1. Cluster centers: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0)$
2. Prob's of clusters: $\pi_{1...K} \sim \text{Dir}(\alpha_{1...K})$
3. Cluster assignments: $z_i \sim \text{Categorical}(\pi_{1...K})$
4. Coordinates of data: $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma_{z_i})$

8.2.2 DP Mixture Model (DP-GMM)

1. Cluster centers: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0)$, $k=1, 2, \dots$
2. Prob's of clusters: $\pi = (\pi_1, \pi_2, \dots) \sim \text{GEM}(\alpha)$
3. Cluster assignments: $z_i \sim \text{Categorical}(\pi)$
4. Coordinates of data: $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma)$, $i=1 \dots N$

Fitting a DP-MM: Collapsed Gibbs sampler
 $p(z_i=k | z_{-i}, x, \alpha, \mu) \propto \text{Prior} \times \text{Likelihood}$
 $\propto \begin{cases} \frac{|x_{-i,k}|}{\alpha+N-1} p(x_i | x_{-i,k}, \mu) & \text{for existing } k \\ \frac{\alpha}{\alpha+N-1} p(x_i | \mu) & \text{otw.} \end{cases}$

$x_{-i,c} := \{x_j | z_j=c, j \neq i\}$ data assigned to clust. c

9 PAC Learning

Want: Distribution indep. error guarantees!

Expec./Gener. error: $\mathcal{R}(\hat{c}_n) = P_{X,Y}(\hat{c}_n(x) \neq c(x))$

Empirical error: $\hat{\mathcal{R}}_n(\hat{c}_n) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{\hat{c}_n(x_i) \neq y_i\}}$

PAC learnable: $\mathcal{A}$ can learn a concept class $\mathcal{C}$ from $\mathcal{H}$ if, given a **suff. large sample**, it outputs a hypothesis that **generalizes well w/ high prob.**

(1) $0 < \epsilon < 1/2$, $0 < \delta < 1/2$, (2) $P_{X,Y}$ on $\mathcal{X} \times \{0, 1\}$:
 If $n \geq \text{poly}(1/\epsilon, 1/\delta, \text{dim}(\mathcal{X}))$,
 Then $P_{X,Y} \left(\mathcal{R}(\hat{c}_n) - \inf_{c \in \mathcal{C}} \mathcal{R}(c) \leq \epsilon \right) \geq 1 - \delta$.

If $\mathcal{A}$ runs in time polynomial in $1/\epsilon$ and $1/\delta$, we say that $\mathcal{C}$ is **efficiently PAC learnable**.

9.1 VC Inequality $P(\dots \geq \epsilon) \leq \dots \leq \delta$

Select ERM: $\hat{c}_n^* = \arg\min_{c \in \mathcal{C}} \hat{\mathcal{R}}_n(c)$
 Under uniform convergence:
 $P(\mathcal{R}(\hat{c}_n^*) - \inf_{c \in \mathcal{C}} \mathcal{R}(c) > \epsilon) \leq P\left(\sup_{c \in \mathcal{C}} |\hat{\mathcal{R}}_n(c) - \mathcal{R}(c)| > \frac{\epsilon}{2}\right)$

- **|C| Finite:** $P(\sup|\dots| > \epsilon) \leq 2|C| \exp(-2n\epsilon^2)$
- **|C| Unbounded:** $P(\dots) \leq 9n^{VC_{\mathcal{C}}} \exp(-\frac{n\epsilon^2}{32})$

9.2 Rectangle Learning

$P((\hat{c}_n^* > \epsilon) \leq |C| \cdot (1-\epsilon)^n \leq |C| \cdot \exp(-n\epsilon) < \delta$
Union bound: $P(\bigcup_i T_i) \leq \sum_i P(T_i)$

10 Appendix

Complete the square: If $p(x) \propto \exp(-\frac{1}{2} x^T A x + x^T b)$, then $p(x) = \mathcal{N}(x | A^{-1} b, A^{-1})$

Constrained optimisation:
primal: $\min_x f(x)$ s.t. $g_i(x) = 0$; $h_j(x) \leq 0$
Lagrangian: with each $\alpha_j \geq 0$
 $\mathcal{L}(x, \lambda, \alpha) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \alpha_j h_j(x)$

Solve: $\frac{\partial \mathcal{L}}{\partial x} = 0$; $g_i(x) = 0$; $\alpha_j \geq 0$; $h_j(x) \leq 0$

If **Slater's cond.** holds, $\exists x: g_i(x) = 0, h_j(x) < 0$, then we can solve the *dual* instead:
 $\max_{\lambda, \alpha} \{\min_x \mathcal{L}(x, \lambda, \alpha)\}$ s.t. $\alpha_j \geq 0$

Solve: $\frac{\partial \mathcal{L}}{\partial x} = 0$; $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$; $\alpha_j h_j(x) = 0$; $\alpha_j \geq 0$

Metrics: $\text{acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN}} \text{ prec} = \frac{\text{TP}}{\text{TP} + \text{FP}} \text{ FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$
 $\text{Recall/TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \text{ balanced acc} = \frac{1}{n} \sum_i \text{TPR}_i$
 $\text{F1-score} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \text{ ROC} = \text{FPR/TPR}$

Conditional Gaussians:
 $P_{X,Y} = \begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right)$, Σ_{ij} p.s.d.
 $\Rightarrow Y | X \sim \mathcal{N}(\tilde{\mu}, \tilde{\Sigma})$, where $\tilde{\mu} = \mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (X - \mu_X)$, $\tilde{\Sigma} = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}$